MSHA Human-Model Validation

Block 4 - Independent Review of 100 Unseen Accident Narratives

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Workbook records: 100
Completed reviews: 100
Rows with at least one uncertain human response: 93

Exact agreement is the primary validation measure.
Exact-or-related agreement is a secondary sensitivity measure recognizing
that short narratives may reasonably fit nearby ontology categories.

Human uncertainty is reported separately and is not silently treated as
model error or model correctness.
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Executive Summary and Interpretation Rules

This report compares the independent human labels in the completed Block 3 workbook with the frozen model output for the same 100 narratives.

The human review is treated as the reference comparison, not as an infallible gold standard. The reviewer explicitly noted that contributing factors and corrective actions were often difficult to determine from the short narratives and that some causes could reasonably fit multiple categories.

The report therefore provides two agreement measures. Exact agreement requires identical ontology labels. Exact-or-related agreement also recognizes predefined neighboring categories that describe similar mechanisms, conditions, or prevention approaches.

UNCERTAIN responses remain visible throughout the report. Certain-only agreement rates exclude records marked uncertain for that field, while all-evaluable rates retain them.

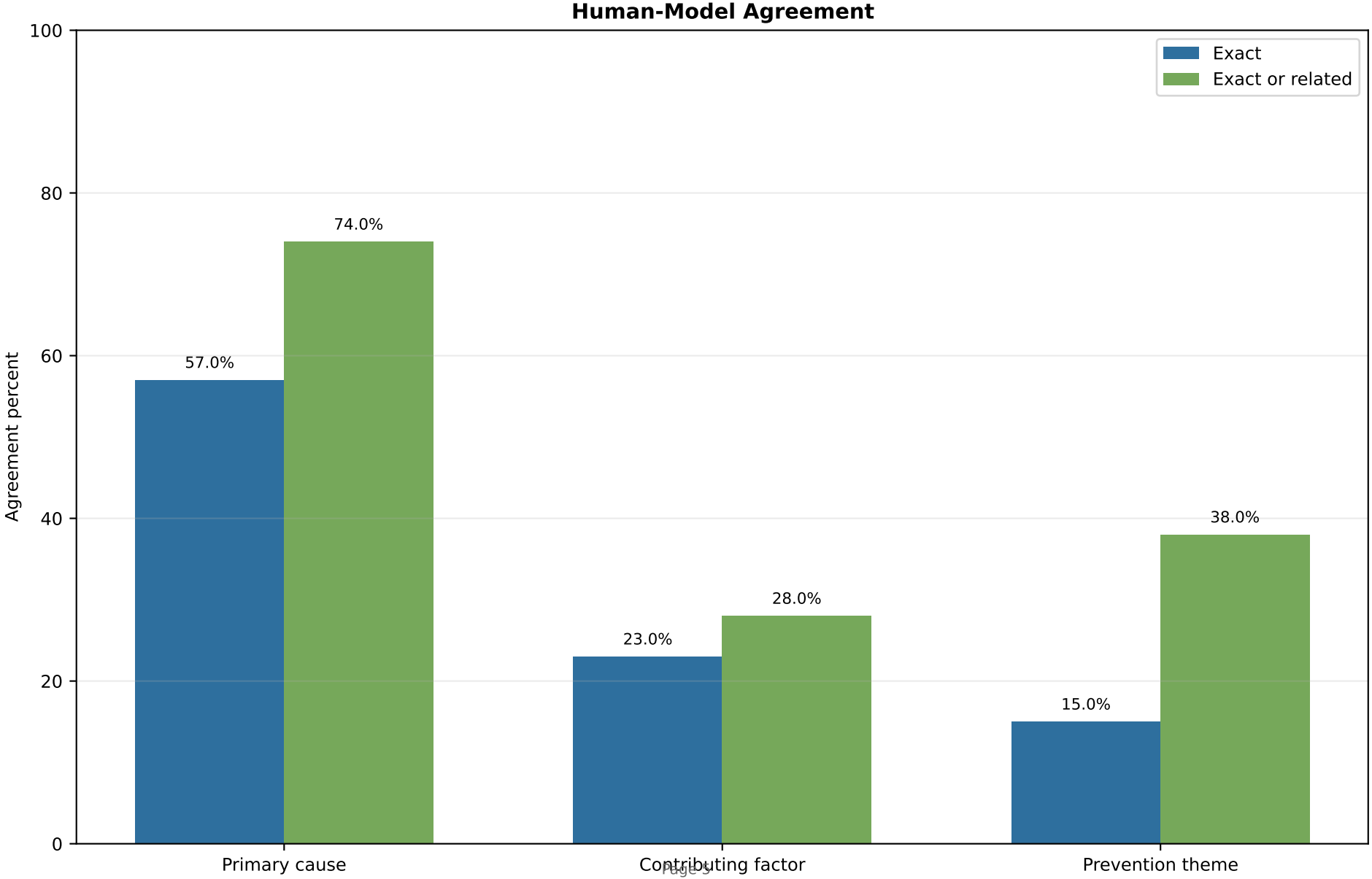
The disagreement examples are intended to support optional reviewer reconsideration. They do not automatically mean that the human or model label is wrong.

Review Completion and Uncertainty

Measure	Value
Workbook rows	100
Review completed = YES	100
Rows with any uncertainty	93
Cause uncertainty	6
Factor uncertainty	77
Prevention uncertainty	89
Human overall acceptable = YES	98
Human overall acceptable = NO	0
Human overall acceptable = UNCERTAIN	2

Agreement Summary

Field	Completed/Evalua	Human Uncertain	Exact Matches	Exact Agreement	Exact Or Related	Exact Or Related	Certain-Only Recc	Certain-Only Exac	Certain-Only Exact O
Primary cause	100	6	57	57.000	74	74.000	94	57.447	75.532
Contributing factor	100	77	23	23.000	28	28.000	23	8.696	17.391
Prevention theme	100	89	15	15.000	38	38.000	11	18.182	27.273


Human-Model Agreement
Exact
Exact or related
Agreement percent
100
80
60
40
20
0
57.0%
74.0%
23.0%
28.0%
15.0%
38.0%
Primary cause
Contributing factor
Prevention theme

Combined Agreement Results

Measure	Count	Percent Of Reviewed Records
Cause and factor both exact	19	19.000
Cause, factor, and prevention all exact	7	7.000
Cause, factor, and prevention exact or related	15	15.000
Human overall acceptable = YES	98	98.000

Aggregate Precision, Recall, F1, and Kappa

Metric	Precision	Recall	F1	Support	Cohen'S Kappa
Cause macro avg	0.524	0.497	0.459	100	0.526
Cause weighted avg	0.645	0.570	0.560	100	0.526
Factor macro avg	0.144	0.153	0.111	100	0.153
Factor weighted avg	0.210	0.230	0.180	100	0.153
Prevention macro avg	0.109	0.116	0.085	100	0.095
Prevention weighted avg	0.138	0.150	0.112	100	0.095

Exact, Related, and Material Disagreement

Field	Exact	Related But Not Exact	Material Disagreement
Primary cause	57	17	26
Contributing factor	23	5	72
Prevention theme	15	23	62

Related-Category Sensitivity Method

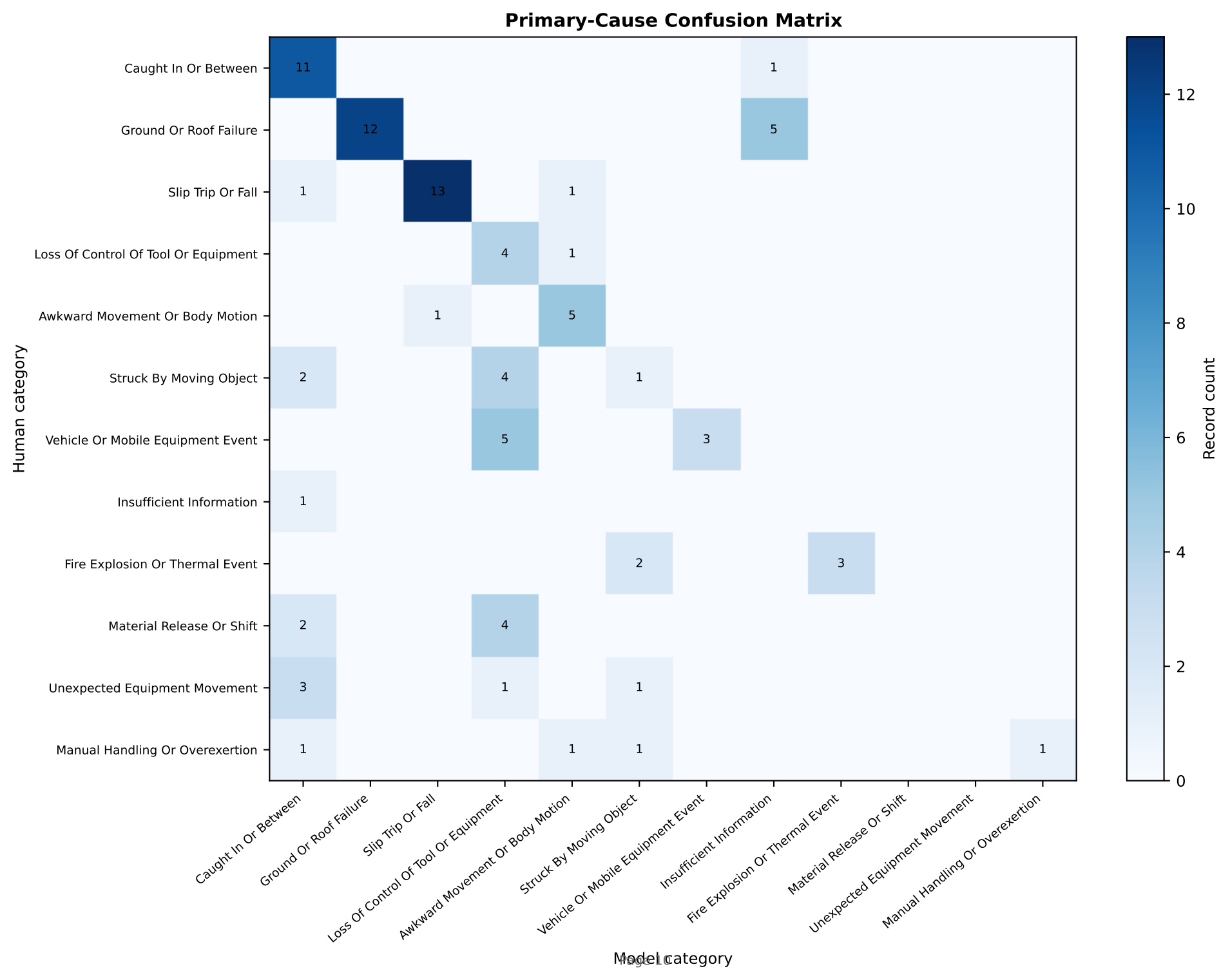
Related matching is a secondary analysis and does not replace exact agreement.

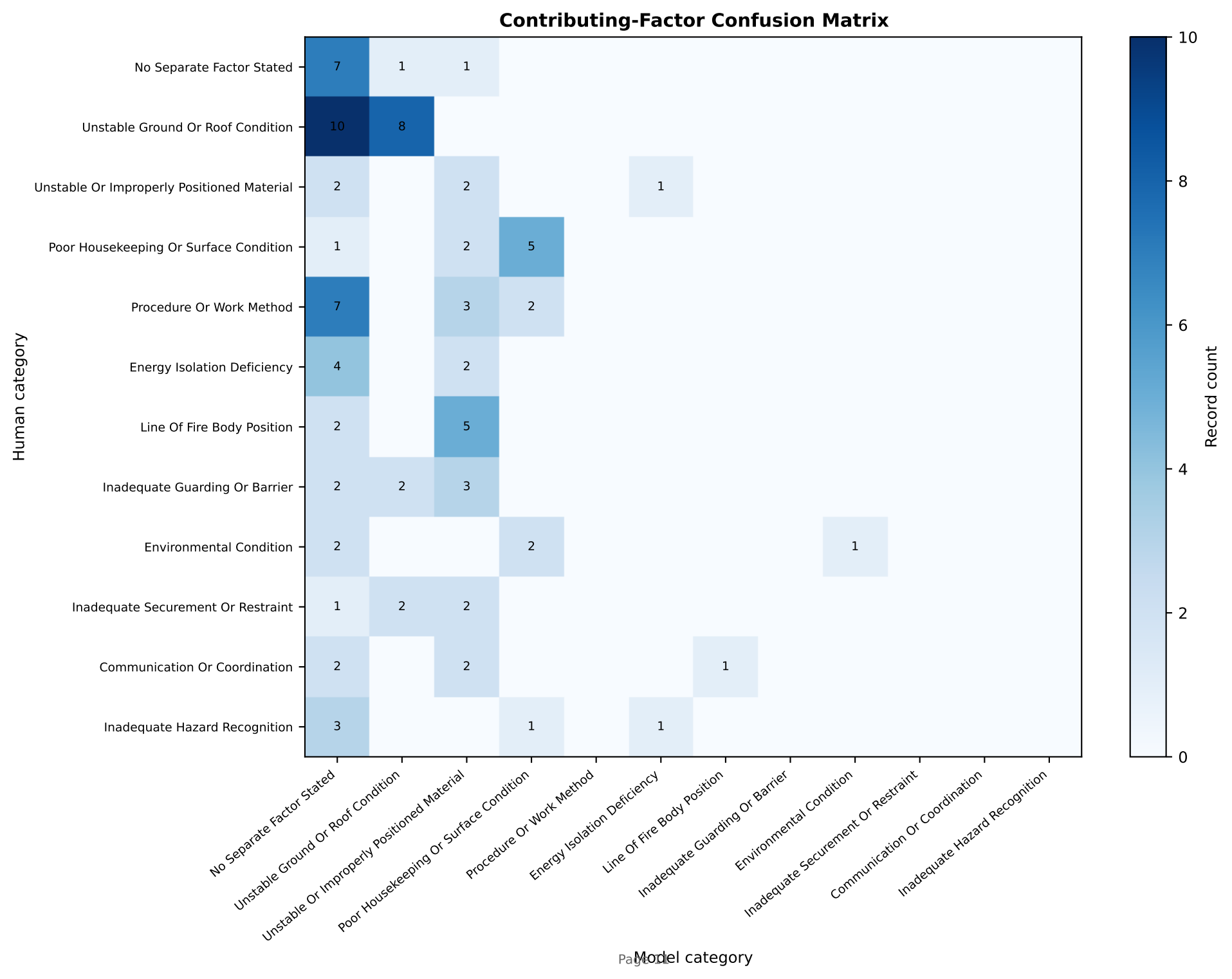
Cause examples include caught-in/between versus struck-by or material release; loss of control versus unexpected equipment movement; and awkward motion versus manual handling or repetitive exposure.

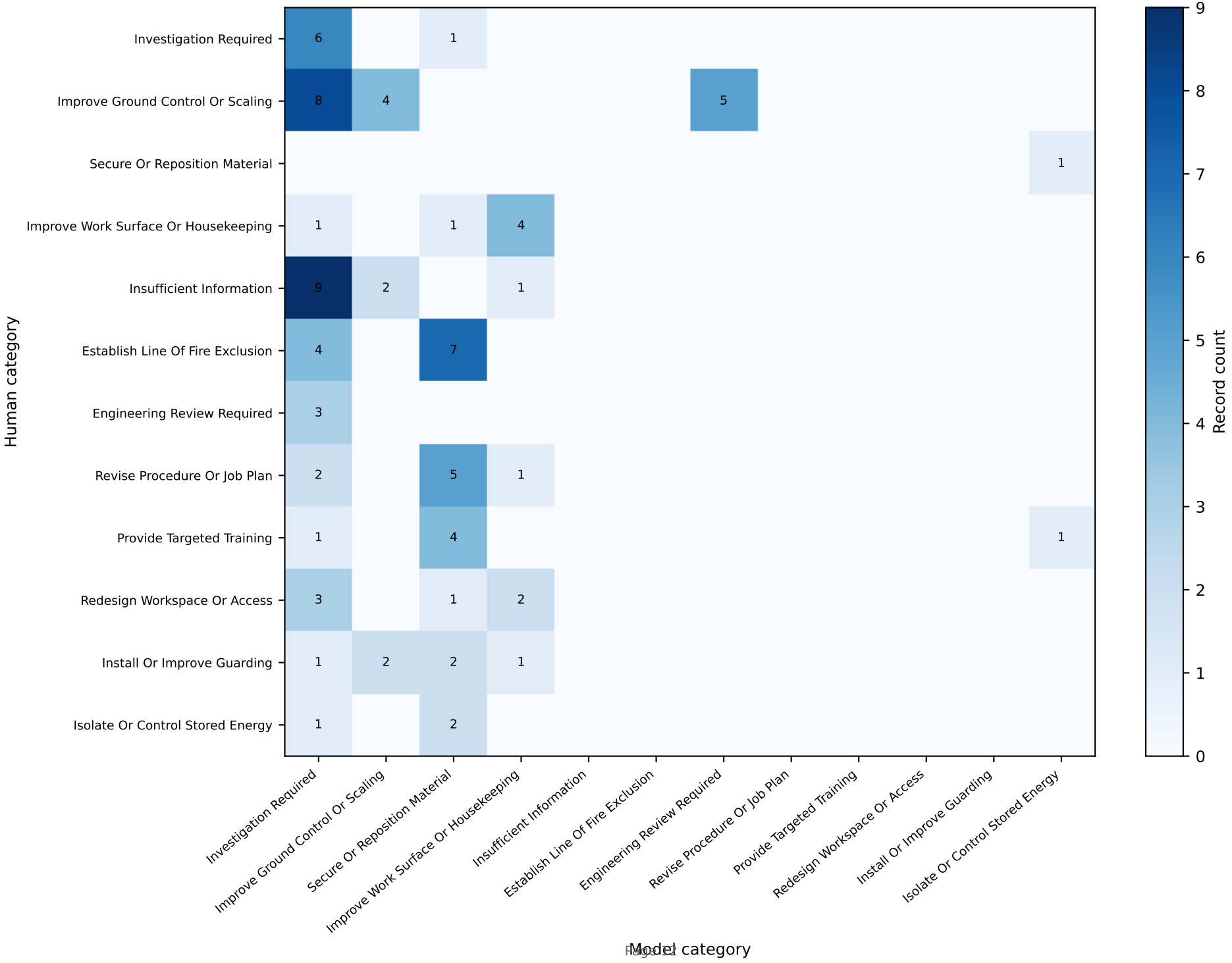
Factor examples include unstable material versus inadequate securement; line-of-fire versus work method or hazard recognition; and equipment condition versus energy isolation or guarding.

Prevention examples include secure/reposition material versus line-of-fire exclusion; equipment repair versus energy isolation or guarding; and procedure revision versus hazard review, training, or communication.

A related match means the two answers are conceptually adjacent within the frozen ontology. It does not establish that both labels are equally correct for the specific narrative.


Primary-Cause Confusion Matrix
Human category
Model category
Record count
Caught In Or Between
Ground Or Roof Failure
Slip Trip Or Fall
Loss Of Control Of Tool Or Equipment
Awkward Movement Or Body Motion
Struck By Moving Object
Vehicle Or Mobile Equipment Event
Insufficient Information
Fire Explosion Or Thermal Event
Material Release Or Shift
Unexpected Equipment Movement
Manual Handling Or Overexertion
0
2
4
6
8
10
12


Contributing-Factor Confusion Matrix
Human category
Model category
Record count
No Separate Factor Stated
Unstable Ground Or Roof Condition
Unstable Or Improperly Positioned Material
Poor Housekeeping Or Surface Condition
Procedure Or Work Method
Energy Isolation Deficiency
Line Of Fire Body Position
Inadequate Guarding Or Barrier
Environmental Condition
Inadequate Securement Or Restraint
Communication Or Coordination
Inadequate Hazard Recognition
0
2
4
6
8
10


Prevention-Theme Confusion Matrix
Human category
Model category
Record count
Investigation Required
Improve Ground Control Or Scaling
Secure Or Reposition Material
Improve Work Surface Or Housekeeping
Insufficient Information
Establish Line Of Fire Exclusion
Engineering Review Required
Revise Procedure Or Job Plan
Provide Targeted Training
Redesign Workspace Or Access
Install Or Improve Guarding
Isolate Or Control Stored Energy
0
1
2
3
4
5
6
7
8
9

Most Common Human-Model Differences

Field	Human Category	Model Category	Count
Primary cause	GROUND_OR_ROOF_FAILURE	INSUFFICIENT_INFORMATION	5
Primary cause	VEHICLE_OR_MOBILE_EQUIPMENT_EVENT	LOSS_OF_CONTROL_OF_TOOL_OR_EQUIPMENT	5
Primary cause	MATERIAL_RELEASE_OR_SHIFT	LOSS_OF_CONTROL_OF_TOOL_OR_EQUIPMENT	4
Primary cause	STRUCK_BY_MOVING_OBJECT	LOSS_OF_CONTROL_OF_TOOL_OR_EQUIPMENT	4
Primary cause	UNEXPECTED_EQUIPMENT_MOVEMENT	CAUGHT_IN_OR_BETWEEN	3
Primary cause	FIRE_EXPLOSION_OR_THERMAL_EVENT	STRUCK_BY_MOVING_OBJECT	2
Primary cause	MATERIAL_RELEASE_OR_SHIFT	CAUGHT_IN_OR_BETWEEN	2
Primary cause	STRUCK_BY_MOVING_OBJECT	CAUGHT_IN_OR_BETWEEN	2
Primary cause	INSUFFICIENT_INFORMATION	HEAT_OR_ENVIRONMENTAL_EXPOSURE	1
Primary cause	INSUFFICIENT_INFORMATION	CAUGHT_IN_OR_BETWEEN	1
Primary cause	GROUND_OR_ROOF_FAILURE	PRESSURE_OR_STORED_ENERGY_RELEASE	1
Primary cause	CHEMICAL_DUST_OR_OCCUPATIONAL_EXPOSURE	FOREIGN_BODY_OR_PARTICLE_CONTACT	1
Primary cause	AWKWARD_MOVEMENT_OR_BODY_MOTION	SLIP_TRIP_OR_FALL	1
Primary cause	CAUGHT_IN_OR_BETWEEN	INSUFFICIENT_INFORMATION	1
Primary cause	LOSS_OF_CONTROL_OF_TOOL_OR_EQUIPMENT	AWKWARD_MOVEMENT_OR_BODY_MOTION	1
Primary cause	MANUAL_HANDLING_OR_OVEREXERTION	AWKWARD_MOVEMENT_OR_BODY_MOTION	1
Primary cause	SLIP_TRIP_OR_FALL	CAUGHT_IN_OR_BETWEEN	1
Primary cause	SLIP_TRIP_OR_FALL	AWKWARD_MOVEMENT_OR_BODY_MOTION	1
Primary cause	MANUAL_HANDLING_OR_OVEREXERTION	CAUGHT_IN_OR_BETWEEN	1
Primary cause	MANUAL_HANDLING_OR_OVEREXERTION	STRUCK_BY_MOVING_OBJECT	1

Cause Category Metrics

Category	Precision	Recall	F1	Human Support	Model Count
AWKWARD_MOVEMENT_OR_BODY	0.625	0.833	0.714	6	8
CAUGHT_IN_OR_BETWEEN	0.524	0.917	0.667	12	21
CHEMICAL_DUST_OR_OCCUPATIO	1.000	0.500	0.667	2	1
ELECTRICAL_CONTACT	1.000	1.000	1.000	1	1
FIRE_EXPLOSION_OR_THERMAL_E	1.000	0.600	0.750	5	3
FOREIGN_BODY_OR_PARTICLE_CO	0.000	0.000	0.000	0	2
GROUND_OR_ROOF_FAILURE	1.000	0.667	0.800	18	12
HEAT_OR_ENVIRONMENTAL_EXPO	0.500	1.000	0.667	1	2
INSUFFICIENT_INFORMATION	0.000	0.000	0.000	2	6
LOSS_OF_CONTROL_OF_TOOL_OF	0.222	0.800	0.348	5	18
MANUAL_HANDLING_OR_OVEREX	1.000	0.250	0.400	4	1
MATERIAL_RELEASE_OR_SHIFT	0.000	0.000	0.000	6	0
PRESSURE_OR_STORED_ENERGY_	0.500	1.000	0.667	1	2
SLIP_TRIP_OR_FALL	0.867	0.867	0.867	15	15
STRUCK_AGAINST_STATIONARY_O	0.000	0.000	0.000	2	0
STRUCK_BY_MOVING_OBJECT	0.200	0.143	0.167	7	5
UNEXPECTED_EQUIPMENT_MOVE	0.000	0.000	0.000	5	0
VEHICLE_OR_MOBILE_EQUIPMENT	1.000	0.375	0.545	8	3

Factor Category Metrics

Category	Precision	Recall	F1	Human Support	Model Count
COMMUNICATION_OR_COORDINA	0.000	0.000	0.000	5	0
ENERGY_ISOLATION_DEFICIENCY	0.000	0.000	0.000	6	2
ENVIRONMENTAL_CONDITION	1.000	0.200	0.333	5	1
EQUIPMENT_CONDITION_OR_FAIL	0.000	0.000	0.000	3	0
INADEQUATE_GUARDING_OR_BAI	0.000	0.000	0.000	7	0
INADEQUATE_HAZARD_RECOGNI	0.000	0.000	0.000	5	0
INADEQUATE_SECUREMENT_OR_I	0.000	0.000	0.000	5	0
LINE_OF_FIRE_BODY_POSITION	0.000	0.000	0.000	7	1
NO_SEPARATE_FACTOR_STATED	0.143	0.778	0.241	9	49
OTHER	0.000	0.000	0.000	2	0
POOR_HOUSEKEEPING_OR_SURFA	0.500	0.625	0.556	8	10
PPE_OR_PROTECTIVE_CONTROL	0.000	0.000	0.000	2	0
PROCEDURE_OR_WORK_METHOD	0.000	0.000	0.000	12	0
UNSTABLE_GROUND_OR_ROOF_C	0.571	0.444	0.500	18	14
UNSTABLE_OR_IMPROPERLY_POS	0.087	0.400	0.143	5	23
WORKSPACE_OR_ACCESS_LIMITA	0.000	0.000	0.000	1	0

Prevention Category Metrics

Category	Precision	Recall	F1	Human Support	Model Count
CONDUCT_PRE_JOB_HAZARD_RE	0.000	0.000	0.000	2	1
CONTROL_ENVIRONMENTAL_EXP	1.000	0.333	0.500	3	1
ENGINEERING_REVIEW_REQUIRE	0.000	0.000	0.000	3	5
ESTABLISH_LINE_OF_FIRE_EXCLU	0.000	0.000	0.000	11	0
IMPROVE_COMMUNICATION_OR_	0.000	0.000	0.000	1	0
IMPROVE_GROUND_CONTROL_OF	0.444	0.235	0.308	17	9
IMPROVE_PPE_OR_PROTECTIVE_E	0.000	0.000	0.000	3	0
IMPROVE_WORK_SURFACE_OR_H	0.400	0.667	0.500	6	10
INSTALL_OR_IMPROVE_GUARDIN	0.000	0.000	0.000	6	0
INSUFFICIENT_INFORMATION	0.000	0.000	0.000	12	0
INVESTIGATION_REQUIRED	0.122	0.857	0.214	7	49
ISOLATE_OR_CONTROL_STORED_	0.000	0.000	0.000	3	2
PROVIDE_TARGETED_TRAINING	0.000	0.000	0.000	6	0
REDESIGN_WORKSPACE_OR_ACC	0.000	0.000	0.000	6	0
REPAIR_REPLACE_OR_INSPECT_E	0.000	0.000	0.000	2	0
REVISE_PROCEDURE_OR_JOB_PLA	0.000	0.000	0.000	8	0
SECURE_OR_REPOSITION_MATER	0.000	0.000	0.000	1	23
USE_MECHANICAL_HANDLING_AS	0.000	0.000	0.000	3	0

Agreement by Severity

Field	Group	Evaluable	Exact %	Exact Or Related %
Cause	Days away	20	55.000	70.000
Cause	Fatality	20	45.000	75.000
Cause	No lost time or other	20	75.000	75.000
Cause	Permanent disability	20	45.000	70.000
Cause	Restricted or transferred	20	65.000	80.000
Factor	Days away	20	25.000	30.000
Factor	Fatality	20	10.000	15.000
Factor	No lost time or other	20	40.000	45.000
Factor	Permanent disability	20	10.000	15.000
Factor	Restricted or transferred	20	30.000	35.000
Prevention	Days away	20	25.000	45.000
Prevention	Fatality	20	15.000	30.000
Prevention	No lost time or other	20	20.000	40.000
Prevention	Permanent disability	20	5.000	25.000
Prevention	Restricted or transferred	20	10.000	50.000

Agreement by Time Period

Field	Group	Evaluable	Exact %	Exact Or Related %
Cause	Early: 2000-2008	35	65.714	80.000
Cause	Middle: 2009-2017	35	60.000	77.143
Cause	Recent: 2018-2026	30	43.333	63.333
Factor	Early: 2000-2008	35	20.000	25.714
Factor	Middle: 2009-2017	35	25.714	34.286
Factor	Recent: 2018-2026	30	23.333	23.333
Prevention	Early: 2000-2008	35	20.000	48.571
Prevention	Middle: 2009-2017	35	14.286	42.857
Prevention	Recent: 2018-2026	30	10.000	20.000

Agreement by Factor-Support Tier

Field	Group	Evaluable	Exact %	Exact Or Related %
Factor	CONTEXTUALLY_SUPPORTED	27	40.741	44.444
Factor	DIRECTLY_SUPPORTED	24	20.833	37.500
Factor	NO_SUPPORTED_FACTOR	49	14.286	14.286
Prevention	CONTEXTUALLY_SUPPORTED	27	25.926	44.444
Prevention	DIRECTLY_SUPPORTED	24	8.333	33.333
Prevention	NO_SUPPORTED_FACTOR	49	12.245	36.735

Agreement by Model Confidence Band

Field	Group	Evaluable	Exact %	Exact Or Related %
Cause	High (>=0.80)	94	60.638	78.723
Cause	Low (<0.65)	6	0.000	0.000
Factor	High (>=0.80)	47	23.404	34.043
Factor	Low (<0.65)	53	22.642	22.642
Prevention	High (>=0.80)	94	15.957	40.426
Prevention	Low (<0.65)	6	0.000	0.000

Cause Differences for Review - 1 of 4

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Example 1 | Record 220033070004 | Fatality | Early: 2000-2008
Status: MATERIAL_DISAGREEMENT
Human category: FIRE_EXPLOSION_OR_THERMAL_EVENT
Model category: STRUCK_BY_MOVING_OBJECT
Related-category group:
Human support assessment: YES
Model confidence: 1.0
Model factor support tier: NO_SUPPORTED_FACTOR
Narrative:
AT 12:30 PM A RIGHT BREAK OFF THE #4 ENTRY SHOT OUT THROUGH AN EXISTING HOLE IN THE RIB OF THE RIGHT SIDE OF THE SHOT INTO THE PARALLEL BREAK, SHOOTING COAL ONTO THE DRILL OPERATOR, DRILL HELP ER AND SHOT FIRER.
Human comments:
nan

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Example 2 | Record 220061300016 | No lost time or other | Early: 2000-2008
Status: MATERIAL_DISAGREEMENT
Human category: GROUND_OR_ROOF_FAILURE
Model category: INSUFFICIENT_INFORMATION
Related-category group:
Human support assessment: YES
Model confidence: 0.0
Model factor support tier: NO_SUPPORTED_FACTOR
Narrative:
Roof fall on the pillar section on 4 Rt.. One break inby spad #3621- measurements: 45'(L) x 20'(W) x 5'(T).
Human comments:
nan

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Example 3 | Record 220070360109 | No lost time or other | Early: 2000-2008
Status: MATERIAL_DISAGREEMENT
Human category: GROUND_OR_ROOF_FAILURE
Model category: INSUFFICIENT_INFORMATION
Related-category group:
Human support assessment: YES
Model confidence: 0.0
Model factor support tier: NO_SUPPORTED_FACTOR
Narrative:
Unintentional roof fall. Approx 80'l x 20'w x 7-10'thick, 40' inby Spad #11,083 at No 6 entry, H intake, at 7 southwest.
Human comments:
nan

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Example 4 | Record 220072670095 | Days away | Early: 2000-2008

Cause Differences for Review - 2 of 4

Status: MATERIAL_DISAGREEMENT
Human category: CHEMICAL_DUST_OR_OCCUPATIONAL_EXPOSURE
Model category: FOREIGN_BODY_OR_PARTICLE_CONTACT
Related-category group:
Human support assessment: NO
Model confidence: 1.0
Model factor support tier: NO_SUPPORTED_FACTOR
Narrative:
EE had went to eat and after he finished he felt discomfort to his right eye. First aid of flushing with water did not help. EE went to doctor on Monday and had a metal sliver removed.
Human comments:
nan

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Example 5 | Record 220072910026 | Fatality | Early: 2000-2008
Status: MATERIAL_DISAGREEMENT
Human category: GROUND_OR_ROOF_FAILURE
Model category: PRESSURE_OR_STORED_ENERGY_RELEASE
Related-category group:
Human support assessment: YES
Model confidence: 1.0
Model factor support tier: NO_SUPPORTED_FACTOR
Narrative:
On August 6, 2007 at 2:50 am a bounce/outburst occurred at the Crandall Canyon Mine. It affected the current mining location in the South Barrier of West Mains extending outby to approx. X/C 119. The outburst affected all 4 entries and rescue operations were begun immediately. Rescue operations to reach the miners were unsuccessful.
Human comments:
nan

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Example 6 | Record 220081890003 | Permanent disability | Early: 2000-2008
Status: MATERIAL_DISAGREEMENT
Human category: UNEXPECTED_EQUIPMENT_MOVEMENT
Model category: CAUGHT_IN_OR_BETWEEN
Related-category group:
Human support assessment: YES
Model confidence: 0.95
Model factor support tier: DIRECTLY_SUPPORTED
Narrative:
While repairing the car body of the radial stacker, the I/E was maneuvering a new section of steel box into place, which was balancing on the radial frame I beam. While re-rigging, the the load shifted causing the I/E end to rise quickly, causing his hand to become caught in a pinch point. The employee suffered the amputation of his left pinky from incident.
Human comments:
nan

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Cause Differences for Review - 3 of 4

Example 7 | Record 220091970012 | Restricted or transferred | Middle: 2009-2017
Status: MATERIAL_DISAGREEMENT
Human category: MATERIAL_RELEASE_OR_SHIFT
Model category: LOSS_OF_CONTROL_OF_TOOL_OR_EQUIPMENT
Related-category group:
Human support assessment: YES
Model confidence: 1.0
Model factor support tier: NO_SUPPORTED_FACTOR
Narrative:
Employee was working on a motor when a pulley slipped and landed on his left hand causing a laceration to his left index finger.
Human comments:
nan

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Example 8 | Record 220092260049 | Days away | Middle: 2009-2017
Status: MATERIAL_DISAGREEMENT
Human category: STRUCK_AGAINST_STATIONARY_OBJECT
Model category: FOREIGN_BODY_OR_PARTICLE_CONTACT
Related-category group:
Human support assessment: YES
Model confidence: 1.0
Model factor support tier: NO_SUPPORTED_FACTOR
Narrative:
EE cut his left index finger, requiring 3 stitches. EE was changing out stainless steel seal segments on the feed end of the kiln, when one of the segments riding (worn) edge cut through his glove and cut his finger.
Human comments:
nan

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Example 9 | Record 220102630059 | No lost time or other | Middle: 2009-2017
Status: MATERIAL_DISAGREEMENT
Human category: GROUND_OR_ROOF_FAILURE
Model category: INSUFFICIENT_INFORMATION
Related-category group:
Human support assessment: YES
Model confidence: 0.0
Model factor support tier: NO_SUPPORTED_FACTOR
Narrative:
A roof fall above anchorage occurred in the No.6 entry adjacent to the belt entry. Fall measured approx 7' X 18' X 20'. 3600 foot outby Jenny branch section.
Human comments:
nan

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Example 10 | Record 220111310033 | Permanent disability | Middle: 2009-2017
Status: MATERIAL_DISAGREEMENT

Cause Differences for Review - 4 of 4

Human category: MANUAL_HANDLING_OR_OVEREXERTION
Model category: CAUGHT_IN_OR_BETWEEN
Related-category group:
Human support assessment: YES
Model confidence: 1.0
Model factor support tier: CONTEXTUALLY_SUPPORTED
Narrative:
Employee was placing 105 lb. Spreader Bar & Dump Block pins into pin hole and the pin slid into place faster than expected crushing employee's left tip of ring finger between the pin head and the dump block..
Human comments:
nan

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Example 11 | Record 220141710003 | Permanent disability | Middle: 2009-2017
Status: MATERIAL_DISAGREEMENT
Human category: MATERIAL_RELEASE_OR_SHIFT
Model category: LOSS_OF_CONTROL_OF_TOOL_OR_EQUIPMENT
Related-category group:
Human support assessment: YES
Model confidence: 1.0
Model factor support tier: NO_SUPPORTED_FACTOR
Narrative:
While repairing a miner waterline rib rolled onto employee. Broken left lower leg & spinal cord injury.
Human comments:
nan

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Example 12 | Record 220150430027 | Days away | Middle: 2009-2017
Status: MATERIAL_DISAGREEMENT
Human category: MATERIAL_RELEASE_OR_SHIFT
Model category: LOSS_OF_CONTROL_OF_TOOL_OR_EQUIPMENT
Related-category group:
Human support assessment: YES
Model confidence: 0.95
Model factor support tier: DIRECTLY_SUPPORTED
Narrative:
EMPLOYEE STATES HE WAS USING A PRY BAR TO UP-RIGHT AND ALIGN THE BALDERSON ATTACHMENT WITH THE FORKS ON A SCOOP. HE PRIED DOWN ON THE BAR, UP-RIGHTING THE ATTACHMENT. ONCE UP-RIGHTED, THE ATTACHMENT TIPPED FORWARD CAUSING THE PRY BAR TO STRIKE THE FRAME OF THE FORKS. THE PRY BAR RAPIDLY SPRANG UPWARD, STRIKING THE EMPLOYEE IN THE HEAD, RESULTING IN A LACERATION.
Human comments:
nan

Contributing-Factor Differences for Review - 1 of 4

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Example 1 | Record 220013600162 | Permanent disability | Early: 2000-2008
Status: MATERIAL_DISAGREEMENT
Human category: PROCEDURE_OR_WORK_METHOD
Model category: UNSTABLE_OR_IMPROPERLY_POSITIONED_MATERIAL
Related-category group:
Human support assessment: YES
Model confidence: 0.95
Model factor support tier: DIRECTLY_SUPPORTED
Narrative:
ATTEMPTING TO REPLACE THE ELEVATOR CHAIN BACK ONTO THE ELEVATOR SPROCKET WHILE IT WAS IN MOTION.HE WAS
USING A BRANCH OFF A CEDAR TREE, WHEN THE CHAIN WAS PRIED ONTO THE SPROCKET THE BRANCH CA UGHT HIS GLOVE
AND PULLED HIS RIGHT HAND BETWEENTHE CHAIN AND SPROCKET.
Human comments:
nan

================================================================================
Example 2 | Record 220022310001 | Fatality | Early: 2000-2008
Status: MATERIAL_DISAGREEMENT
Human category: INADEQUATE_GUARDING_OR_BARRIER
Model category: UNSTABLE_GROUND_OR_ROOF_CONDITION
Related-category group:
Human support assessment: YES
Model confidence: 0.0
Model factor support tier: CONTEXTUALLY_SUPPORTED
Narrative:
EMPLOYEE STEPPED OFF 23 FT HIGHWALL ONTO ROCKS WAS STILL CONSCIENCE AFTER THE FALL, WAS TAKEN BY
AMBULANCE TO KITTANNING HOSPITAL.
Human comments:
nan

================================================================================
Example 3 | Record 220040290010 | Permanent disability | Early: 2000-2008
Status: MATERIAL_DISAGREEMENT
Human category: ENERGY_ISOLATION_DEFICIENCY
Model category: UNSTABLE_OR_IMPROPERLY_POSITIONED_MATERIAL
Related-category group:
Human support assessment: YES
Model confidence: 1.0
Model factor support tier: DIRECTLY_SUPPORTED
Narrative:
IN AN ATTEMPT TO UNCLOG AUGER ON 6064 MACK POWDER TRUCK THE EE REMOVED THE OR COVER PLATE AND PLACE HIS
LEFT HAND INTO THE ROTARING AUGER. RESULTING IN AN AMPUTARIONT TO THE MIDDLE FINGER OF T HE LEFT HAND
BETWEEN THE FIRST TWO JOINTS AND LACERATION TO THE RING AND INDEX FINGER OF THE LEFT HAND.
Human comments:
nan

================================================================================
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Contributing-Factor Differences for Review - 2 of 4

Example 4 | Record 220045550003 | Fatality | Early: 2000-2008
Status: MATERIAL_DISAGREEMENT
Human category: INADEQUATE_SECUREMENT_OR_RESTRAINT
Model category: UNSTABLE_GROUND_OR_ROOF_CONDITION
Related-category group:
Human support assessment: YES
Model confidence: 0.95
Model factor support tier: DIRECTLY_SUPPORTED
Narrative:
A FATAL FALL OF ROOF ACCIDENT OCCURRED ON THE 003 WORKING SECTION. THE VICTIM WAS WORKING WITH THE SECTION CREW AND HAD BEEN INSTALLING TIMBERS DURING RETREAT MINING. THE CREW HAD TEMPORARILY HALTED PRODUCTION TO OBSERVE THE MINE ROOF, WHEN WITH LITTLE WARNING THE MINE ROOF BEGAN TO COLLAPSE. THE 5 MINERS RAN TO ESCAPE THE ROOF FALL. FOUR OF THE MINERS ESCAPED.
Human comments:
nan

==
Example 5 | Record 220050590068 | Permanent disability | Early: 2000-2008
Status: MATERIAL_DISAGREEMENT
Human category: PROCEDURE_OR_WORK_METHOD
Model category: UNSTABLE_OR_IMPROPERLY_POSITIONED_MATERIAL
Related-category group:
Human support assessment: YES
Model confidence: 1.0
Model factor support tier: DIRECTLY_SUPPORTED
Narrative:
Hooking up pull chain to dogs. Caught fingers between dog & pull chain. Crushed middle, index & ring finger.
Human comments:
nan

==
Example 6 | Record 220072120012 | Days away | Early: 2000-2008
Status: MATERIAL_DISAGREEMENT
Human category: INADEQUATE_SECUREMENT_OR_RESTRAINT
Model category: UNSTABLE_GROUND_OR_ROOF_CONDITION
Related-category group:
Human support assessment: YES
Model confidence: 0.95
Model factor support tier: CONTEXTUALLY_SUPPORTED
Narrative:
The injured was in the process of retrieving a B.O. Getman tractor out of the slope. In the process, the tractor started sliding and struck the arches in the slope causing an avulsion to the back of his head.
Human comments:
nan

==

Contributing-Factor Differences for Review - 3 of 4

Example 7 | Record 220072670095 | Days away | Early: 2000-2008
Status: MATERIAL_DISAGREEMENT
Human category: PPE_OR_PROTECTIVE_CONTROL
Model category: NO_SEPARATE_FACTOR_STATED
Related-category group:
Human support assessment: NO
Model confidence: 0.0
Model factor support tier: NO_SUPPORTED_FACTOR
Narrative:
EE had went to eat and after he finished he felt discomfort to his right eye. First aid of flushing with water did not help. EE went to doctor on Monday and had a metal sliver removed.
Human comments:
nan

==
Example 8 | Record 220072910026 | Fatality | Early: 2000-2008
Status: MATERIAL_DISAGREEMENT
Human category: UNSTABLE_GROUND_OR_ROOF_CONDITION
Model category: NO_SEPARATE_FACTOR_STATED
Related-category group:
Human support assessment: YES
Model confidence: 0.0
Model factor support tier: NO_SUPPORTED_FACTOR
Narrative:
On August 6, 2007 at 2:50 am a bounce/outburst occurred at the Crandall Canyon Mine. It affected the current mining location in the South Barrier of West Mains extending outby to approx. X/C 119. The outburst affected all 4 entries and rescue operations were begun immediately. Rescue operations to reach the miners were unsuccessful.
Human comments:
nan

==
Example 9 | Record 220080390028 | Permanent disability | Early: 2000-2008
Status: MATERIAL_DISAGREEMENT
Human category: COMMUNICATION_OR_COORDINATION
Model category: UNSTABLE_OR_IMPROPERLY_POSITIONED_MATERIAL
Related-category group:
Human support assessment: YES
Model confidence: 0.95
Model factor support tier: CONTEXTUALLY_SUPPORTED
Narrative:
Two emloyes were manually rotating a light mast in the dark. While rotating the mast one employee had his left hand on the handle and his right hand on the mast in the pinch point, the rotation of the mast severed the tips of his middle and ring fingers on his right hand between the rotating mast and mast support.
Human comments:
nan

Contributing-Factor Differences for Review - 4 of 4

==========
Example 10 | Record 220122690039 | Days away | Middle: 2009-2017
Status: MATERIAL_DISAGREEMENT
Human category: POOR_HOUSEKEEPING_OR_SURFACE_CONDITION
Model category: UNSTABLE_OR_IMPROPERLY_POSITIONED_MATERIAL
Related-category group:
Human support assessment: YES
Model confidence: 0.8
Model factor support tier: CONTEXTUALLY_SUPPORTED
Narrative:
The injured was setting some cribs for supplemental support. As he was beating a wedge over top of the crib the hammer caught some lifeline and caused him to miss the wedge and hit some fingers.
Human comments:
nan

==========
Example 11 | Record 220132050032 | Days away | Middle: 2009-2017
Status: MATERIAL_DISAGREEMENT
Human category: PROCEDURE_OR_WORK_METHOD
Model category: NO_SEPARATE_FACTOR_STATED
Related-category group:
Human support assessment: NO
Model confidence: 0.0
Model factor support tier: NO_SUPPORTED_FACTOR
Narrative:
Operator was attempting to climb back up and into his Komatsu 830, as operator reached for the handrail felt a pop in left shoulder area.
Human comments:
nan

==========
Example 12 | Record 220160890010 | Fatality | Middle: 2009-2017
Status: MATERIAL_DISAGREEMENT
Human category: WORKSPACE_OR_ACCESS_LIMITATION
Model category: NO_SEPARATE_FACTOR_STATED
Related-category group:
Human support assessment: YES
Model confidence: 0.0
Model factor support tier: NO_SUPPORTED_FACTOR
Narrative:
On March 22, 2016 at approximately 6:35pm an employee was struck by flyrock after a shot was detonated. The employee was sitting inside of a pickup approximately 1,200 feet away from the shot when detonated. This incident resulted in a fatality.
Human comments:
nan

Prevention-Theme Differences for Review - 1 of 4

```
================================================================================
Example 1 | Record 220000450001 | Fatality | Early: 2000-2008
Status: MATERIAL_DISAGREEMENT
Human category: IMPROVE_GROUND_CONTROL_OR_SCALING
Model category: INVESTIGATION_REQUIRED
Related-category group:
Human support assessment: YES
Model confidence: 0.9
Model factor support tier: NO_SUPPORTED_FACTOR
Narrative:
ON JAN. 31, 2000, AT APPROXIMATELY 8:00 PM, A FALL OF GROUND OCCURRED IN THE #215 STOPE AREA THAT CAUSED
A FATAL INJURY, INJURY TO OTHER PERSONNEL AND PROPERTY DAMAGE. EXACT SEQUENCE OF EVENTS ARE STILL
UNKNOWN AND UNDERGOING INVESTIGATION.
Human comments:
nan

================================================================================
Example 2 | Record 220022310001 | Fatality | Early: 2000-2008
Status: MATERIAL_DISAGREEMENT
Human category: INSTALL_OR_IMPROVE_GUARDING
Model category: IMPROVE_GROUND_CONTROL_OR_SCALING
Related-category group:
Human support assessment: YES
Model confidence: 1.0
Model factor support tier: CONTEXTUALLY_SUPPORTED
Narrative:
EMPLOYEE STEPPED OFF 23 FT HIGHWALL ONTO ROCKS WAS STILL CONSCIENCE AFTER THE FALL, WAS TAKEN BY
AMBULANCE TO KITTANNING HOSPITAL.
Human comments:
nan

================================================================================
Example 3 | Record 220032250036 | Days away | Early: 2000-2008
Status: MATERIAL_DISAGREEMENT
Human category: ESTABLISH_LINE_OF_FIRE_EXCLUSION
Model category: INVESTIGATION_REQUIRED
Related-category group:
Human support assessment: YES
Model confidence: 1.0
Model factor support tier: NO_SUPPORTED_FACTOR
Narrative:
EE HAD JUST ARRIVED ON SECTION HE WALKED UP BETWEEN THE MINER AND THE RIB WHEN THE RIB ROLLED HITTING
HIM ON HIS LEFT LEG.
Human comments:
nan

================================================================================
Example 4 | Record 220033070004 | Fatality | Early: 2000-2008
```

Prevention-Theme Differences for Review - 2 of 4

Status: MATERIAL_DISAGREEMENT
Human category: ESTABLISH_LINE_OF_FIRE_EXCLUSION
Model category: INVESTIGATION_REQUIRED
Related-category group:
Human support assessment: YES
Model confidence: 1.0
Model factor support tier: NO_SUPPORTED_FACTOR
Narrative:
AT 12:30 PM A RIGHT BREAK OFF THE #4 ENTRY SHOT OUT THROUGH AN EXISTING HOLE IN THE RIB OF THE RIGHT SIDE OF THE SHOT INTO THE PARALLEL BREAK, SHOOTING COAL ONTO THE DRILL OPERATOR, DRILL HELP ER AND SHOT FIRER.
Human comments:
nan

==========
Example 5 | Record 220122690039 | Days away | Middle: 2009-2017
Status: MATERIAL_DISAGREEMENT
Human category: INVESTIGATION_REQUIRED
Model category: SECURE_OR_REPOSITION_MATERIAL
Related-category group:
Human support assessment: YES
Model confidence: 0.9
Model factor support tier: CONTEXTUALLY_SUPPORTED
Narrative:
The injured was setting some cribs for supplemental support. As he was beating a wedge over top of the crib the hammer caught some lifeline and caused him to miss the wedge and hit some fingers.
Human comments:
nan

==========
Example 6 | Record 220160890010 | Fatality | Middle: 2009-2017
Status: MATERIAL_DISAGREEMENT
Human category: REDESIGN_WORKSPACE_OR_ACCESS
Model category: INVESTIGATION_REQUIRED
Related-category group:
Human support assessment: YES
Model confidence: 1.0
Model factor support tier: NO_SUPPORTED_FACTOR
Narrative:
On March 22, 2016 at approximately 6:35pm an employee was struck by flyrock after a shot was detonated. The employee was sitting inside of a pickup approximately 1,200 feet away from the shot when detonated. This incident resulted in a fatality.
Human comments:
nan

==========
Example 7 | Record 220170410014 | Permanent disability | Middle: 2009-2017
Status: MATERIAL_DISAGREEMENT

Prevention-Theme Differences for Review - 3 of 4

Human category: REDESIGN_WORKSPACE_OR_ACCESS
Model category: SECURE_OR_REPOSITION_MATERIAL
Related-category group:
Human support assessment: YES
Model confidence: 0.9
Model factor support tier: DIRECTLY_SUPPORTED
Narrative:
Miner climbed up stockpile to observe conditions of head roll on pea rock conveyor. Miner lost footing and right hand went through drive pulley pinch point, slicing three fingers through miners glove.
Human comments:
nan

==========
Example 8 | Record 220192110010 | Fatality | Recent: 2018-2026
Status: MATERIAL_DISAGREEMENT
Human category: ISOLATE_OR_CONTROL_STORED_ENERGY
Model category: INVESTIGATION_REQUIRED
Related-category group:
Human support assessment: YES
Model confidence: 0.0
Model factor support tier: NO_SUPPORTED_FACTOR
Narrative:
Workers removing a jaw crusher motor. As the hammer attachment was being positioned towards the crusher motor, the hydraulic hammer attachment fell from the excavator quick coupler. The hammer attachment struck the victim in the head pining them to the work platform.
Human comments:
nan

==========
Example 9 | Record 220011030031 | Restricted or transferred | Early: 2000-2008
Status: DISAGREEMENT_UNCERTAIN
Human category: USE_MECHANICAL_HANDLING_ASSISTANCE
Model category: INVESTIGATION_REQUIRED
Related-category group:
Human support assessment: UNCERTAIN
Model confidence: 0.8
Model factor support tier: NO_SUPPORTED_FACTOR
Narrative:
EE WAS BENT OVER TO WRAP CHAIN AROUND A TILT CYLINDER AND HE STRAINED HIS LOWER BACK.
Human comments:
nan

==========
Example 10 | Record 220011970077 | Days away | Early: 2000-2008
Status: DISAGREEMENT_UNCERTAIN
Human category: CONDUCT_PRE_JOB_HAZARD_REVIEW
Model category: INVESTIGATION_REQUIRED
Related-category group:

Prevention-Theme Differences for Review - 4 of 4

Human support assessment: UNCERTAIN
Model confidence: 1.0
Model factor support tier: NO_SUPPORTED_FACTOR
Narrative:
EE WAS CLEANING UP AROUND FEEDER IN CLINKER TUNNEL AND GOT DUST IN EYES. HIS EYES WERE FLUSHED AT THE PLANT BUT IRRITATION CONTINUED. EE WAS SEEN BY AN EYE DR. AND RECEIVED A PRESCRIPTION FOR EYE DROPS.
Human comments:
nan

==========
Example 11 | Record 220013400047 | Days away | Early: 2000-2008
Status: DISAGREEMENT_UNCERTAIN
Human category: ESTABLISH_LINE_OF_FIRE_EXCLUSION
Model category: INVESTIGATION_REQUIRED
Related-category group:
Human support assessment: UNCERTAIN
Model confidence: 1.0
Model factor support tier: NO_SUPPORTED_FACTOR
Narrative:
THE WATERLINE CAME LOOSE FROM THE SHEAR STRIKINGHIM ON THE RIGHT HAND/WRIST.
Human comments:
nan

==========
Example 12 | Record 220013600162 | Permanent disability | Early: 2000-2008
Status: DISAGREEMENT_UNCERTAIN
Human category: PROVIDE_TARGETED_TRAINING
Model category: SECURE_OR_REPOSITION_MATERIAL
Related-category group:
Human support assessment: UNCERTAIN
Model confidence: 0.95
Model factor support tier: DIRECTLY_SUPPORTED
Narrative:
ATTEMPTING TO REPLACE THE ELEVATOR CHAIN BACK ONTO THE ELEVATOR SPROCKET WHILE IT WAS IN MOTION.HE WAS USING A BRANCH OFF A CEDAR TREE, WHEN THE CHAIN WAS PRIED ONTO THE SPROCKET THE BRANCH CA UGHT HIS GLOVE AND PULLED HIS RIGHT HAND BETWEENTHE CHAIN AND SPROCKET.
Human comments:
nan

Conclusions and Appropriate Use

Primary-cause performance should be interpreted from exact agreement, exact-or-related agreement, category-level F1, and the cause confusion matrix together.

Contributing-factor and prevention performance should be interpreted more cautiously because many narratives describe what happened without providing enough detail to establish why it happened or which corrective action is best.

The factor-support-tier analysis tests whether directly supported model factors perform better than contextually supported factors. Confidence-band results test whether the model's numerical confidence corresponds to higher human agreement.

Records marked uncertain should not be used as definitive evidence for or against a category. They are retained as a transparent part of the validation set and may be updated after reviewing the disagreement examples.

If the human workbook is revised, rerun this Block 4 script. The report and comparison outputs will be regenerated from the updated labels without rerunning the language model.

Technical Appendix

```
Input workbook: /mnt/batch/tasks/shared/LS_root/mounts/clusters/dvandyk2/code/Users/dvandyk/NW Data
Science/MSDS 453- AI and Natural Language/Final Project/MSHA_Block3_Human_Review_Workbook-
Completed.xlsx
Records compared: 100
Expected records: 100
Completed reviews: 100
Rows with uncertainty: 93
Comparison CSV: /mnt/batch/tasks/shared/LS_root/mounts/clusters/dvandyk2/code/Users/dvandyk/NW Data
Science/MSDS 453- AI and Natural Language/Final Project/MSHA_Block4_Human_Model_Comparison.csv
Differences workbook: /mnt/batch/tasks/shared/LS_root/mounts/clusters/dvandyk2/code/Users/dvandyk/NW
Data Science/MSDS 453- AI and Natural Language/Final Project/MSHA_Block4_Disagreements_for_Review.xlsx
PDF report: /mnt/batch/tasks/shared/LS_root/mounts/clusters/dvandyk2/code/Users/dvandyk/NW Data
Science/MSDS 453- AI and Natural Language/Final Project/MSHA_Block4_Human_Validation_Report.pdf

Exact agreement is primary. Related agreement depends on the explicit related-category groups defined at
the beginning of this script.
```